How Tip Options Shape Tipping Decisions: A Randomized Survey Simulation

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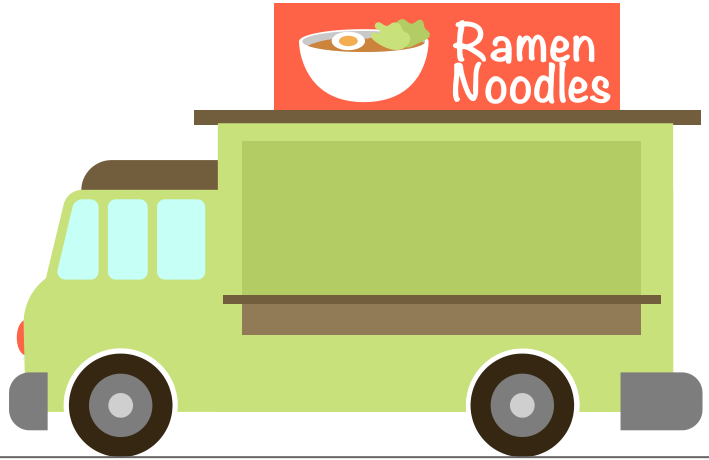
Research Question:

How does showing higher tip suggestions affect the tip percentage selected in a simulated dine-in restaurant checkout?

H_0

There is no effect of showing higher default tip suggestions on the tip percentage selected.



H_a

There is an effect of showing higher default tip suggestions on the tip percentage selected.

Experimental Design

Platform

Qualtrics online survey

Scenario

scripted dining experience;
fixed bill **$16 Tonkotsu ramen**

Participants

classmates + friends

Flow

scenario → checkout tip
choice → short
post-questions

Final sample

N = 81 (after removing
preview + incomplete)

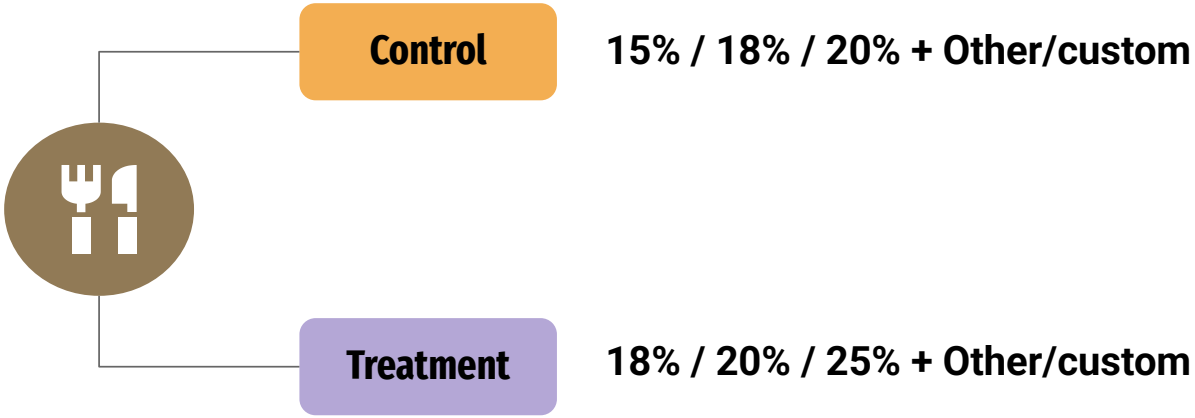
Pilot

7–10 respondents;
minor clarity edits

Treatment & Randomization

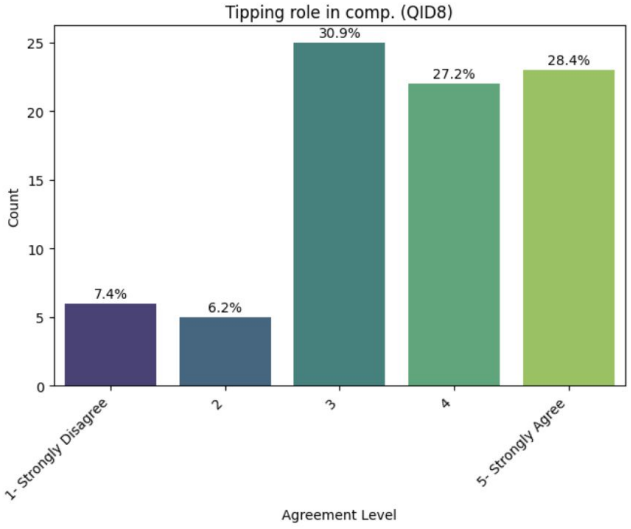
Manipulation

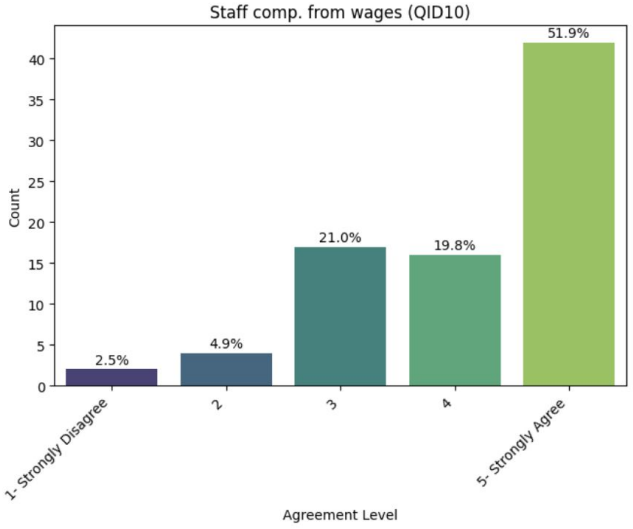
Default tip options
on payment screen



Randomization

- **Unit:** individual respondent
- **Method:** Qualtrics built-in randomizer (intended ~50/50)
- **Timing:** assigned immediately before the payment screen





Cleaning

- Created **Treatment indicator** and merged control/treatment survey versions into a unified dataset
- Missing values handled by **listwise deletion when required**

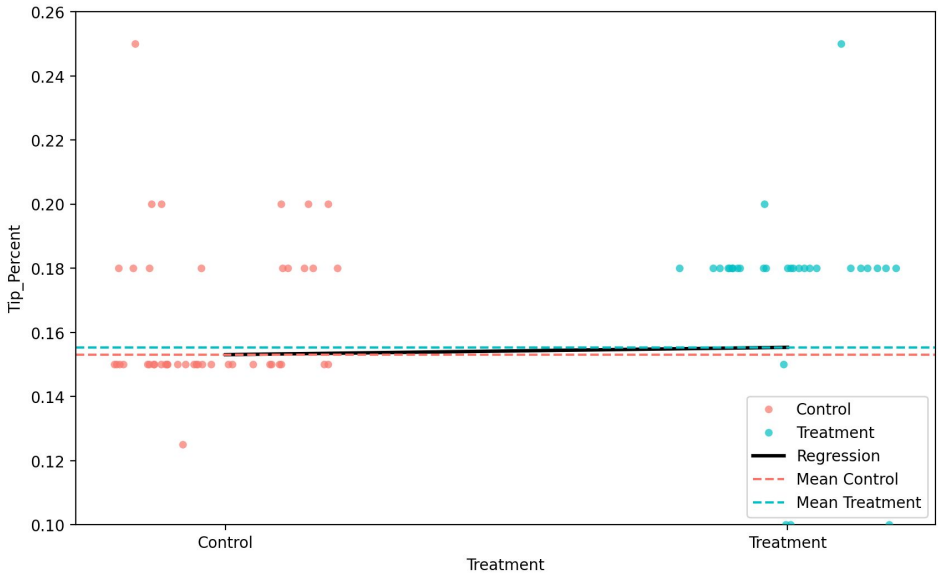
EDA

- Demographic and behavior charts: **age, race, years in U.S., dining frequency**
- Attitude results: respondents **recognize tipping's role**, but many also **support wage-based compensation**
- Survey duration check: most responses **20–180s**; **<20s treated as inattentive**

Randomization Check

- Groups: **47 control (58%) vs 34 treatment (42%)**
- **χ^2 test p = 0.149** → no significant imbalance
- Covariates balanced (**Age p = 0.837, Years in U.S. p = 0.440**)

Methods & Results



Regression: Effect of Treatment on Tip_Percent

	Dependent variable: Tip_Percent
	(1)
	(1)
Intercept	0.153***
	(0.007)
Treatment	0.002
	(0.012)
Observations	81
R^2	0.000
Adjusted R^2	-0.012
Residual Std. Error	0.052 (df=79)
F Statistic	0.035 (df=1; 79)
Note:	*p<0.1; **p<0.05; ***p<0.01

- Linear Regression detects 0.2 percentage point increase in tipping percentage for treatment vs. control
- **SE** = 1.2 percentage point indicates insignificant results

Methods & Results

Regression: Treatment Effect on Tip_Percent (with covariates)

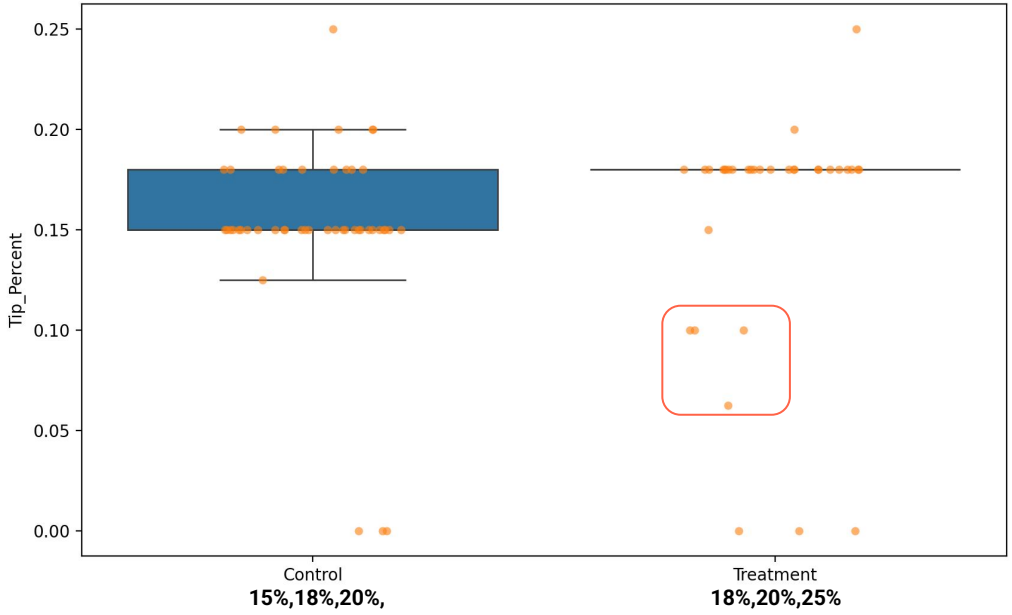
	Dependent variable: Tip_Percent
	(1)
	(1)
Age: 25–40	-0.000
	(0.015)
Age: <18	-0.024
	(0.042)
Age: Prefer not	-0.030**
	(0.014)
US years: ≥5	0.028**
	(0.011)
US years: <1	-0.019
	(0.017)
Service industry: Yes	-0.006
	(0.012)
Dining: 4–7/wk	-0.012
	(0.020)
Dining: almost never	0.006
	(0.018)
Dining: >7/wk	0.014
	(0.014)
Intercept	0.152***
	(0.009)
Treatment	0.008
	(0.012)
Observations	81

Heterogeneity Analysis: Effect of Treatment on Tip_Percent

	Dependent variable: Tip_Percent
	(1)
	(1)
Intercept	0.173***
	(0.034)
Wages should cover pay	-0.004
	(0.005)
US ≥ 5 years	0.042**
	(0.016)
Tipping funds pay	-0.005
	(0.007)
Treatment	0.063
	(0.056)
Treatment × Wages should cover pay	-0.019
	(0.014)
Treatment × US ≥ 5 years	0.007
	(0.022)
Treatment × Tipping funds pay	0.006
	(0.011)
Observations	81

- Covariate Analysis:
  - Age, Service-Industry Experience, Dining Frequency, and **Years Lived in the U.S**.
- Heterogeneity Analysis:
  - Years Lived in the U.S,
  - “Tipping plays a role in how restaurant workers are compensated”(1-5)
  - “Restaurant staff compensation should primarily come from wages paid by employers”(1-5)

Interpretation

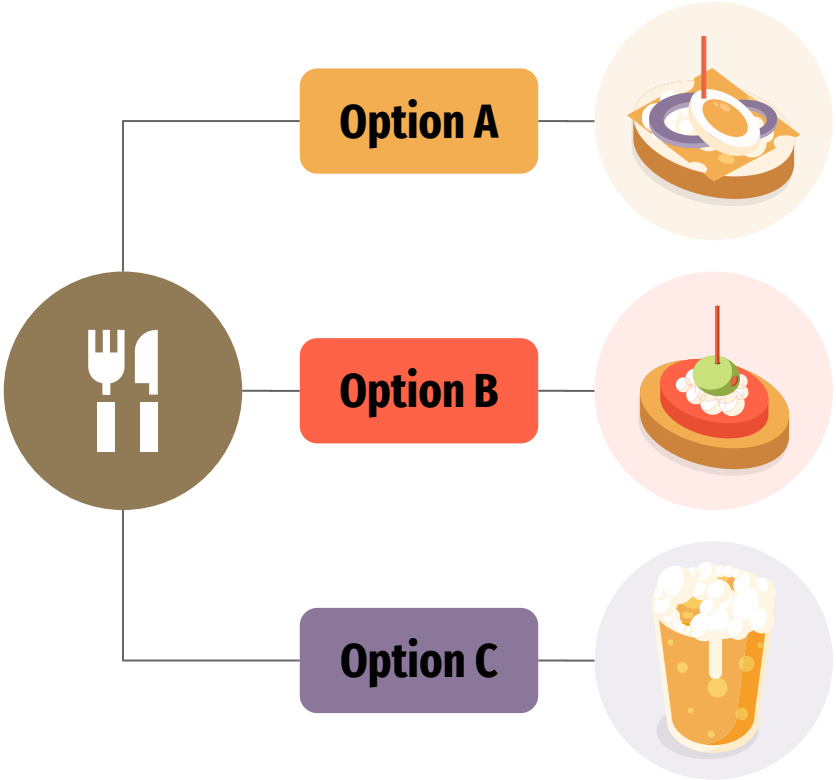


- Participants usually pick the lowest preset tip option.
- Treatment (18/20/25): most choose 18%; few choose 20% or 25%.
- Raising the minimum to 18% shifts some 15% choosers up to 18%.
- Others use “customise” to tip below 18% (sometimes very low), offsetting the increase.
- 25% may be too high; a smaller top option (e.g., 22%) could encourage more tips.

Business Insight

Result

Simply increasing the available tipping percentage from 15% to 18% will have a **limited impact** on tipping behavior.



Emphasize the **importance** of tips to employees income.

Restaurants could run **small**-scale **A/B tests** on their own payment systems.

Provides **multiple design** options for the tips selection interface

Limitation

01

Sample Bias

Most of our data were collected from BU student.

02

Fake payment

People are not actually paying money for tips

03

Simulated scenario difference

People may interpret the simulated scenario differently, which could lead them to choose different tip amounts.

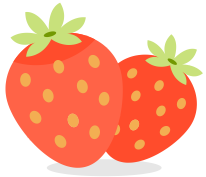
04

Limited sample size

The sample size we have right now may not be able to find small differences

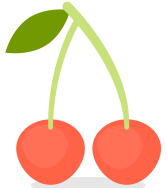
Conclusions

Higher default tip suggestions did not significantly increase tipping



Tipping choices remain anchored around social norms (15–18%)

Interface design and behavioral nudges may be more effective than changing defaults



Future A/B testing in real payment systems can identify better tipping strategies

Thank You!
Any Questions?